

UNIVERSIDAD TÉCNICA PARTICULAR DE LOJA

ANÁLISIS DE ALGORITMOS

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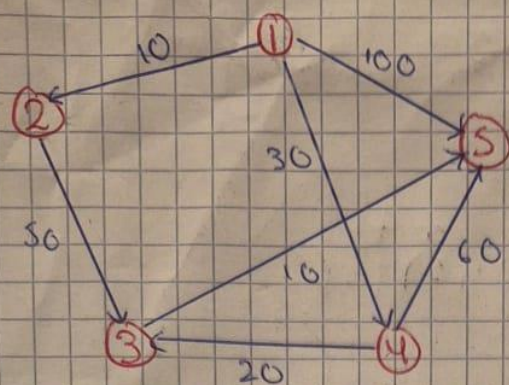
Taller

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Fecha: 09-06-2025

$N = \{1, 2, 3, 4, 5, 6\}$

$A = \{ \{1, 2\}, \{1, 3\}, \{1, 4\}, \{2, 3\}, \{2, 5\}, \{3, 4\}, \{3, 5\}, \{3, 6\}, \{4, 6\}, \{5, 6\} \}$

T	R	$N \setminus R$	$e = \{u, v\}$	$ R $
$\{1, 3\}$	$\{1, 3\}$	$2, 4, 5, 6$	$\{1, 3\}$	1
$\{1, 3\}$	$\{1, 3, 6\}$	$2, 4, 5$	$\{3, 6\}$	4
$\{1, 3\}$	$\{1, 3, 6, 4\}$	$2, 5$	$\{6, 4\}$	2
$\{1, 3\}$	$\{1, 3, 6, 4, 5\}$	2	$\{2, 5\}$	3
$\{1, 3\}$	$\{1, 3, 6, 4, 5, 2\}$	$\emptyset$	$\{3, 2\}$	5
$\{1, 3\}, \{3, 6\}$				15
$\{6, 4\}, \{2, 5\}$				
$\{3, 2\}$				



	1	2	3	4	5
1	-	10	-	30	100
2	-	-	50	-	-
3	-	-	-	-	10
4	-	-	20	-	60
5	-	-	-	-	-

Posos	u	c	S	D (vertices)			
1	1	{2,3,4,5}	1	10	-	30	100
	2	{3,4,5}	1,2	-	60	30	100
	4	{3,5}	1,2,4	-	50	-	80
	3	5	1,2,4,3	-	60	-	60

$$D[u,v] = \min(D[u,w], D[v,w] + d[u,w])$$

$$D[2] = \min(10, 100 + \infty)$$

$$D[3] = \min(\infty, 10 + 30)$$

$$D[4] = \min(30, 10 + \infty)$$

$$D[5] = \min(100, 10 + \infty)$$

$$D[3] = \min(60, 30 + 20)$$

$$D[5] = \min(100, 30 + 60)$$

$$D[5] = \min(80, 50 + 10)$$

